A uniform lower bound for the Faltings height of Jacobians of Shimura curves X_0^D(M)

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Abstract

Let $J_0^D(M)$ be the Jacobian of the Shimura curve $X_0^D(M)$ attached to an indefinite quaternion algebra of discriminant $D > 1$ over $\mathbf{Q}$ and an Eichler order of level M, and let g be its genus. For $D = 1$, Jorgenson and Kramer proved $h_F(J_0(N)) \sim \frac{g}{3}\log N$ for squarefree N prime to 6; for $D > 1$ the size of the Faltings height in terms of the level has been open. We prove that if $N = DM$ is squarefree and prime to 6 and $g \geq 2$, then $12\,h_F(J_0^D(M)) \geq 4(g-1)\big[\sum_{q\mid D}\frac{q}{q-1}\log q + \sum_{p\mid M}\frac{p}{p+1}\log p\big] - C\big(g + 2^{\omega(N)}\log N\big)$ with an absolute constant C, hence $h_F(J_0^D(M)) \geq (\frac{1}{3} - o(1))\,g\log N$ uniformly in (D, M). The proof applies the arithmetic Riemann-Roch theorem of Freixas i Montplet and von Pippich for orbifold curves. The self-intersection of the orbifold canonical bundle is computed from Yuan's modular height formula and an error-free level-raising recursion on the normal coarse model, in which thick nodes enter through Mumford intersection numbers and the Eichler mass formula. We also show that elliptic sections are disjoint at primes ≥ 5 and have bounded intersections at 2 and 3, compute their heights, and give a one-sided bound for the Selberg zeta value. As a consequence, inside an explicitly stated exponent-profile relaxation model that retains Pasten's separate bound $v_2(abc) \ll_\epsilon R^\epsilon$, the Petersson-ratio channel of Pasten's Shimura-curve approach to abc cannot lower the Stewart-Yu exponent 1/3 on an infinite family of triples whose Frey curves have conductor prime to 6; this obstruction is forced by the height bound, not by the model. We claim no upper bound or asymptotic for the height, and nothing about the abc conjecture itself. The manuscript was produced with AI research agents and has not been reviewed by a human expert.

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Problem and contribution

Let B be the indefinite quaternion algebra over $\mathbf{Q}$ of discriminant $D > 1$, let $M \geq 1$ be coprime to D, put $N = DM$, and let $J = J_0^D(M)$ be the Jacobian of the Shimura curve $X = X_0^D(M)$ of genus g. For $D = 1$ (the modular curves $X_0(N)$ with N squarefree and coprime to 6), Jorgenson and Kramer [10, Theorem 6.2] proved $h_F(J_0(N)) = \frac{g}{3}\log N + o(g\log N)$. For $D > 1$ the size of $h_F(J_0^D(M))$ in terms of the level has remained open; Pasten [16, Section 1] records that its computation or asymptotic evaluation is not known. The existing results relate it to other quantities only modulo $\mathbf{Q}$-linear combinations of $\log p$, $p \mid DM$ [18, 6], and those are exactly the terms of size $g\log N$.

Unconditional result (Theorem 1.1). Assume $N = DM$ is squarefree, $\gcd(N, 6) = 1$, $D > 1$ and $g \geq 2$. Then

$$12\,h_F(J_0^D(M)) \;\geq\; 4(g-1)\Big[\sum_{q\mid D}\frac{q}{q-1}\log q + \sum_{p\mid M}\frac{p}{p+1}\log p\Big] - C\big(g + 2^{\omega(N)}\log N\big)$$

with an absolute constant C. The factor $q/(q-1) > 1$ at the primes of D records the full contribution of the Čerednik–Drinfeld fibres. In particular $h_F(J_0^D(M)) \geq (\frac{1}{3} - o(1))\,g\log N$, uniformly in the pair (D, M).

Model-relative consequence (Corollary 7.4). In the explicit relaxation model of Definition 7.1, which retains Pasten's separate estimate $v_2(abc) \ll_\epsilon R^\epsilon$ [16, Lemma 15.2, (56)], the Petersson-ratio method of [16] cannot lower the Stewart–Yu exponent 1/3 [20] on an infinite family of abc triples, namely triples whose Frey curve has conductor prime to 6 (for instance $(16t, 1, 16t+1)$ with $t \equiv 1 \pmod 6$ and t divisible by sufficiently many primes). The obstruction comes from Proposition 7.3, that is, from Theorem 1.1: if the product bounds were of size $N^{o(1)}$, then for each fixed $\epsilon > 0$ the choice $C_2 = C'_\epsilon R^\epsilon$ would give $\log c \le R^{\epsilon+o(1)}$ in the same model, and since such a bound at 2 is available for every $\epsilon > 0$, the model would then give subpower bounds (Remark 7.5). This is a statement about one channel of one method inside a stated model; it is not a statement about the abc conjecture.

Not claimed. No upper bound and no asymptotic formula for $h_F(J_0^D(M))$; nothing for $\gcd(N, 6) > 1$ or for $D = 1$; no improvement or obstruction for abc beyond Definition 7.1; no claim about Pasten's full argument, which also uses linear forms in logarithms, or about the congruence-module argument of [27].

The constant 1/3 coincides with the Jorgenson–Kramer constant for $D = 1$, so the theorem is consistent with the expectation $h_F(J_0^D(M)) \sim \frac{g}{3} \log N$.

Method in one paragraph

We apply the arithmetic Riemann–Roch theorem of Freixas i Montplet and von Pippich for orbifold curves with elliptic fixed points [7, Theorem 10.1] to a regular model of X over $\mathbf{Z}[i, \zeta_3]$. The main term is the self-intersection of the orbifold canonical bundle $\bar{L}_M = K + \sum(1 - 1/m_i)\sigma_i$. At level $M = 1$ it is given by Yuan's modular height formula [25, Theorem 1.1]. We show that raising the level by a prime p changes $\bar{L}^2$ by an explicit amount with no error term (Proposition 4.4). Two inputs make this possible: we work on the normal coarse model, and we evaluate thick nodes through Mumford's rational intersection numbers and the Eichler mass formula. The remaining terms are bounded separately. The finite intersections of elliptic sections vanish at $p \ge 5$ and are $O(1)$ per section at 2, 3 (Section 3). The heights of elliptic sections are $\frac{1}{2} \log N + O(1)$ (Section 5). An elementary heat-trace argument gives a one-sided bound on the Selberg zeta value $\log Z'(1)$ (Lemma 6.1).

Relation to prior work

For modular curves the self-intersection $\bar{\omega}^2$ of the relative dualizing sheaf and the Faltings height of $J_0(N)$ are known asymptotically: Abbes–Ullmo [1, Théorèmes A, B, Proposition C] and Michel–Ullmo [15, Théorème 1.1] for $\bar{\omega}^2$ on $X_0(N)$, Ullmo [21, Théorème 1.2] for bounds on $h_F(J_0(N))$, Jorgenson–Kramer [10, Theorem 6.2] for the asymptotic, and further $X_1(N)$ [13, Theorem 7.7], $X(N)$ [9, Theorem 18], $X_0(p^2)$ [2, Theorem 1.2]. All of these use cusps (Rankin–Selberg unfolding, Eisenstein series, or covers of a fixed base curve), and none of these tools is available on the compact curves $X_0^D(M)$. For Shimura curves, Kudla–Rapoport–Yang [12], Kühn [11] (for $D = 1$) and Yuan [25, Theorem 1.1] computed the height of the Hodge bundle, i.e. $\bar{L}^2$ at maximal level. That is an invariant of the Hodge bundle, not of the Jacobian. Prasanna [18] and Freixas i Montplet [6] related $h_F(J_0^D(M))$ to modular-curve invariants through the Jacquet–Langlands correspondence, in both cases only up to rational multiples of $\log p$, $p \mid DM$. Pasten [16, Section 1] states that the asymptotic evaluation for quaternionic Shimura curves is open. We did not find in the literature a lower bound of order $g \log N$ for $h_F(J_0^D(M))$ with $D > 1$. The elementary bound with constant 1/12 in place of 1/3 (Remark 6.3) follows from Faltings' Noether formula, Faltings' positivity theorem and Wilms' bound for the delta invariant. The present paper replaces positivity of $\bar{\omega}^2$ by an evaluation of $\bar{L}^2$.

Notation and conventions

Throughout, $N = DM$ is squarefree, $\gcd(N, 6) = 1$, and $D > 1$ is a product of an even number of primes. Write $\varphi(D) = \prod_{q \mid D}(q-1)$, $\psi(M) = \prod_{p \mid M}(p+1)$ and $d_M = \varphi(D)\psi(M)/6$. The curve $X(\mathbf{C}) = \Gamma \backslash \mathfrak{H}$ is compact. It has e_2 elliptic points of order 2 and e_3 of order 3, with $e_2, e_3 \le 2^{\omega(N)}$ (classical formulas of Shimura and Ogg), and $e = e_2 + e_3$. By the Riemann–Hurwitz formula for orbifolds,

$$d_M = 2g - 2 + \frac{1}{2}e_2 + \frac{2}{3}e_3 \;\ge\; 2g - 2. \tag{0.1}$$

Every elliptic point has CM by $\mathbf{Z}[i]$ or $\mathbf{Z}[\zeta_3]$ and is defined over $K = \mathbf{Q}(i, \zeta_3)$. All arithmetic degrees and intersection numbers are divided by $[K : \mathbf{Q}] = 4$. For a finite place v of K and sections σ, τ meeting at a closed point we write $(\sigma, \tau)_v = \frac{1}{4} \operatorname{length}(\mathcal{O}/(I_\sigma + I_\tau)) \log N(v)$, and $(\sigma, \tau)_{\mathrm{fin}} = \sum_v (\sigma, \tau)_v$. "$O(\cdot)$" constants are absolute.

1. Statement

Theorem 1.1. There is an absolute constant C such that for all D, M as above with $g \geq 2$,

$$12\, h_F(J_0^D(M)) \;\geq\; 4(g-1)\Big[\sum_{q\mid D} \frac{q}{q-1}\log q + \sum_{p\mid M}\frac{p}{p+1}\log p\Big] - C\big(g + 2^{\omega(N)}\log N\big).$$

Here h_F is the Faltings height of J. Since N is squarefree, J has semistable reduction over $\mathbf{Q}$, so the height over $\mathbf{Q}$ and the stable height coincide.

Corollary 1.2. $h_F(J_0^D(M)) \geq (\frac{1}{3} - o(1))\, g \log N$ as $N \to \infty$, uniformly in (D, M).

Proof of 1.2 from 1.1. Since $\frac{q}{q-1} > 1$, the bracket in Theorem 1.1 is at least $\log D + \sum_{p\mid M}\frac{p}{p+1}\log p$, and $\log D + \sum_{p\mid M}\frac{p}{p+1}\log p = \log N - \sum_{p\mid M}\frac{\log p}{p+1}$. For any $Y \geq 1$, $\sum_{p\mid M}\frac{\log p}{p+1} \leq \sum_{p\leq Y}\frac{\log p}{p+1} + \frac{\log N}{Y+1}$; letting $N \to \infty$ and then $Y \to \infty$ shows that this sum is $o(\log N)$ uniformly. Since $d_M = \varphi(D)\psi(M)/6 \geq \varphi(N)/6$, the genus formula (0.1) gives $g \geq d_M/2 - O(2^{\omega(N)}) \gg N/\log\log N$, while $2^{\omega(N)}\log N = N^{o(1)}$; hence $C(g + 2^{\omega(N)}\log N) = o(g \log N)$. □

2. Models

2.1. The normal coarse models. Let $\mathcal{X}_1$ be the coarse moduli scheme over $\mathbf{Z}$ attached to a maximal order of B. This is the normal integral model X_U for maximal U used in [25, Section 1.1] and [26, Section 2]. For $M \geq 1$ let $\mathcal{X}_M$ be the normalization of $\mathcal{X}_1$ in the function field of $X_0^D(M)_{\mathbf{Q}}$. It is a normal, flat, projective $\mathbf{Z}$-surface. Each forgetful map $\pi : \mathcal{X}_M \to \mathcal{X}_{M/p}$ $(p \mid M)$ is finite. The surface $\mathcal{X}_M$ coincides with the coarse moduli scheme of the $\Gamma_0(M)$-moduli problem. Indeed, choose an auxiliary level $\ell \geq 5$ prime to N at which the moduli problem is representable. Buzzard [3, Theorem 4.7] shows that the resulting fine model with $\Gamma_0(p)$-structure is regular and finite over the fine model without it. The coarse model is the quotient by a finite group, and the formation of such quotients commutes with flat base change and with completion [4, Theorems 1.1 and 3.1]. The coarse model is therefore normal and finite over $\mathcal{X}_1$, and by uniqueness of normalization it equals $\mathcal{X}_M$. The same argument run over $\mathbf{Z}[1/\ell]$ for two different auxiliary primes covers all of $\operatorname{Spec}\mathbf{Z}$.

2.2. Reduction. We use the following facts from Stankewicz [19]. - If $p \nmid N$, then $\mathcal{X}_M \otimes \mathbf{F}_p$ is smooth [19, Theorem 2.8]. This includes $p = 2, 3$; Lemma 3.3 below gives a direct proof at $2, 3$. - If $p \mid M$, the special fibre is the union of two copies C_1, C_2 of $\mathcal{X}_{M/p} \otimes \mathbf{F}_p$, meeting at the superspecial points. These points are in bijection with the right ideal classes $[I]$ of an Eichler order of level M/p in the definite algebra B_{Dp} of discriminant Dp. The completed strict local ring at the point attached to $[I]$ is $W[[X,Y]]/(XY - p^{\ell([I])})$ with $\ell([I]) = \#\mathcal{O}_l(I)^\times/\{\pm 1\}$ [19, Theorem 2.8, Lemma 2.9, Definition 2.5]. The forgetful map π restricts to the identity on one component and to the Frobenius on the other [19, Theorem 2.8]. We call these C_1 (identity) and C_2 (Frobenius). - If $q \mid D$, the Čerednik–Drinfeld uniformization gives a Mumford curve over $\mathbf{Z}_{q^2}$ whose components are indexed by two copies of the classes of level M in $B_{D/q}$. The double points are indexed by the classes of level Mq in $B_{D/q}$, with local rings $\mathbf{Z}_{q^2}[[X,Y]]/(XY - q^{\ell([I])})$ [19, Theorem 2.6].

2.3. The orbifold canonical bundle. Let $\sigma_1, \ldots, \sigma_e$ be the Zariski closures over $\mathcal{O}_K$ of the elliptic points, $m_i \in \{2, 3\}$ their orders, $a_i = 1 - 1/m_i$. Put

$$L_M = K_{\mathcal{X}_M/\mathbf{Z}} + \sum_i a_i \sigma_i,$$

a $\mathbf{Q}$-Weil divisor class on the normal surface $\mathcal{X}_M$. At $M = 1$ this is Yuan's Hodge bundle: on each open set where an auxiliary level U' is maximal, L_1 equals $\omega + \sum(1 - e_Q^{-1})Q$ by [25, Section 1.1], and the pull-back of L_1 to fine levels is the relative dualizing sheaf [26, Lemma 2.1]. On $X(\mathbf{C})$ we give L_M the orbifold hyperbolic metric, normalized as in [25] by $\|f(\tau)d\tau\| = 2\operatorname{Im}(\tau)|f(\tau)|$. Changing to another constant multiple of this metric (for instance the curvature -1 normalization of [7]) changes $\bar{L}_M^2$ by $O(d_M)$. The metric pulls back isometrically under the maps π, which are unramified as maps of orbifolds.

Intersection numbers on the normal surfaces $\mathcal{X}_M$ are Mumford's rational intersection numbers: a Weil divisor is pulled back to the minimal resolution by the unique $\mathbf{Q}$-divisor that is orthogonal to the exceptional curves. The projection formula $\pi^* A \cdot B = A \cdot \pi_* B$ holds.

3. Elliptic sections

Lemma 3.1. Let R be an order in a definite quaternion algebra over $\mathbf{Q}$ containing non-commuting elements u, v with $\mathbf{Z}[u], \mathbf{Z}[v]$ isomorphic to $\mathbf{Z}[i]$ or $\mathbf{Z}[\zeta_3]$. Then the discriminant of the algebra divides 2, 3 or 4; in particular it has no prime factor ≥ 5. Consequently, every finite subgroup of $R^\times/\{\pm 1\}$ is cyclic of order 1, 2 or 3 when the discriminant has a prime factor ≥ 5.

Proof. Let $d_u, d_v \in \{-4, -3\}$ be the discriminants of u, v and put $x = 2\,\mathrm{trd}(uv) - \mathrm{trd}(u)\mathrm{trd}(v)$. The order $\mathbf{Z}\langle 1, u, v, uv\rangle \subseteq R$ has reduced discriminant $(d_u d_v - x^2)/4$. This is positive because u, v do not commute, and $x \equiv d_u d_v \pmod 2$. The possible values are $\{3,4\}$, $\{2,3\}$, $\{2\}$ for $(d_u, d_v) = (-4,-4), (-4,-3), (-3,-3)$. The discriminant of the algebra divides the reduced discriminant of every order. For the second assertion: a non-cyclic finite subgroup of $R^\times/\{\pm 1\}$ contains two non-commuting elements of order 2 or 3, whose lifts generate quadratic orders of discriminant -4 or -3. □

Proposition 3.2. Let $i \neq j$ and let v be a place of K above a prime $p \geq 5$. Then $(\sigma_i, \sigma_j)_v = 0$. Moreover, at $p \mid N$ every σ_i meets the special fibre in a smooth point.

Proof. Suppose σ_i, σ_j pass through the same closed point x of the fibre at p, and let k_i, k_j be their CM fields.

Case $p \nmid N$. If x is supersingular, the $\mathcal{O}_B$-linear endomorphism algebra of the corresponding abelian surface is the definite algebra B_{Dp} of discriminant Dp, so by Lemma 3.1 the two CM orders commute; if x is ordinary, this endomorphism algebra is an imaginary quadratic field and hence already commutative. In either case the two CM orders generate the same maximal order $\mathcal{O}_k$ of one imaginary quadratic field k, with the same image, and p is inert in k when x is supersingular and splits in k when x is ordinary. By Morita equivalence [3, Theorem 4.5, Corollary 4.6], deforming the abelian surface with its $\mathcal{O}_B$-action and prime-to-p level structure is equivalent to deforming one p-divisible group G of height 2 and dimension 1. The locus in the deformation space where the action of $\mathcal{O}_k \otimes \mathbf{Z}_p$ lifts is the deformation space of G as a formal $\mathcal{O}_k \otimes \mathbf{Z}_p$-module. If p is inert in k, then G is a formal $\mathcal{O}_{k,p}$-module of height 1. Such a module has a unique lift once the residual action is fixed, namely the canonical lifting [8, Proposition 2.1]; the residual action on the Lie algebra distinguishes the two embeddings of $\mathcal{O}_k$. If x is ordinary, so p splits in k, the Serre–Tate canonical lift is the unique lift with the given action. In both cases the two sections coincide in the formal neighbourhood of x, hence everywhere, which is a contradiction.

Case $p \mid M$. A $\Gamma_0(p)$-structure stable under $\mathcal{O}_k$ is, after Morita reduction, an $\mathcal{O}_k/p$-stable line in $\mathbf{F}_p^2$. If p were inert in k, then $\mathcal{O}_k/p \cong \mathbf{F}_{p^2}$ would have no stable line; hence p splits in k and x is ordinary. Ordinary points are smooth points of the special fibre (they are not superspecial), and the argument of the previous case applies verbatim. Pasten [16, Lemma 13.1] records the same smoothness for Heegner points.

Case $p \mid D$. Then p is inert in k, since k embeds in B and $p \nmid \mathrm{disc}(k)$. In the Čerednik–Drinfeld uniformization $\widehat{\mathcal{X}} \otimes W \cong \Gamma' \backslash \widehat{\Omega} \otimes W$, with Γ' an arithmetic group in the definite algebra B' of discriminant D/p, an elliptic point is the image of a fixed point z of a torsion element $\gamma \in \Gamma'$. Since $\mathbf{Q}_p(\gamma)$ is the unramified quadratic extension, z and $\bar{z}$ reduce into the open part of one vertex component $\mathbf{P}^1_s$, at the two conjugate points of $\mathbf{P}^1(\mathbf{F}_{p^2}) \setminus \mathbf{P}^1(\mathbf{F}_p)$. In particular the reduction is a smooth point. If σ_i, σ_j meet at x, choose lifts z_i, z_j reducing to the same point of the same vertex component s. The corresponding torsion elements γ_i, γ_j lie in the vertex stabilizer, which is $R_s^\times/\{\pm 1\}$ for an order R_s of B'. Here D/p has a prime factor ≥ 5, so by Lemma 3.1 this group is cyclic. Hence γ_i, γ_j have the same fixed points, so $z_j \in \{z_i, \bar{z}_i\}$. Since $\bar{z}_i$ reduces to the conjugate point, which is different, $z_j = z_i$ and $\sigma_i = \sigma_j$. □

Lemma 3.3. Let $p \in \{2, 3\}$. (a) $\mathcal{X}_M \otimes \mathbf{Z}_p$ is smooth. (b) Two distinct elliptic sections meet at a place $v \mid p$ only if both have CM by $\mathcal{O}_k$ with $(k, p) = (\mathbf{Q}(i), 2)$ or $(\mathbf{Q}(\zeta_3), 3)$. At most two elliptic sections pass through a given closed point, and $(\sigma_i, \sigma_j)_2 \leq 2\log 2$, resp. $(\sigma_i, \sigma_j)_3 \leq \frac{3}{2}\log 3$. (c) $\sum_{i \neq j} a_i a_j (\sigma_i, \sigma_j)_{\mathrm{fin}} \leq 2e_2 \log 2 + \frac{3}{2} e_3 \log 3 = O(e)$.

Proof. (a) Off the points with non-trivial automorphisms the coarse and fine models agree locally and are smooth [3, Theorem 4.7(i)]. Let x be a point whose effective automorphism group $H = \mathrm{Aut}(x)/\{\pm 1\}$ is non-trivial. Since $p \nmid D$ and D has a prime factor ≥ 5, Lemma 3.1 shows that H is cyclic of order $m \in \{2, 3\}$, generated by the image of a CM unit. The completed local ring of the fine deformation problem (after Morita reduction) is $A = W[[t]]$, and the coarse completed local ring is A^H [4, Theorem 3.1]. Put $z = \prod_{h \in H} h(t)$. An automorphism of $W[[t]]$ need not send t to t times a unit over W (the involution of Remark 3.4 has $h(0) = -2$). Modulo p, however, the reduction of h is a local automorphism of $k[[t]]$, where k is the residue field of W, so $h(t) \equiv t \cdot (\text{unit}) \pmod p$, and hence $z \equiv (\text{unit}) \cdot t^m \pmod p$. This is all that the Weierstrass argument uses. By the Weierstrass preparation theorem, A is free of rank m

over $W[[z]]$. Since H acts faithfully on the generic fibre, $[\mathrm{Frac}(A) : \mathrm{Frac}(A)^H] = m$, and $W[[z]] \subseteq A^H$ has the same fraction field as A^H. As $W[[z]]$ is integrally closed, $A^H = W[[z]]$. This argument does not require $p \nmid m$.

(b) If the CM types differ, the two orders do not commute and Lemma 3.1 (with discriminant Dp) excludes this. If p is unramified in k, the argument of Proposition 3.2 gives disjointness. In the ramified case, fix the residual action. By the theory of canonical liftings [8, Proposition 2.1], the locus $Z \subset \mathrm{Spf}\, W[[t]]$ where the $\mathcal{O}_k \otimes \mathbf{Z}_p$-action lifts is $\mathrm{Spf}(W \otimes_{\mathbf{Z}_p} \mathcal{O}_{k,p}) = \mathrm{Spf}\, W[\varpi]$. Here ϖ is a uniformizer of k_p, and t restricts to a generator θ of $W[\varpi]$ over W. Its two geometric points are θ and its conjugate $\bar{\theta}$. Hence at most two elliptic sections pass through x, and their lifts satisfy $v_\varpi(t_1 - t_2) = v_\varpi(\theta - \bar{\theta}) = v_\varpi(\mathfrak{d}_{k_p/\mathbf{Q}_p})$. This is 2 for $\mathbf{Q}_2(i)$ and 1 for $\mathbf{Q}_3(\zeta_3)$, in agreement with [23, Section 4]. The group H is generated by a unit of $\mathcal{O}_k$ and acts on Z through automorphisms of the canonical lift, so it fixes t_1 and t_2. Since $Q(t) = \prod_{h \in H}(h(t) - t_2)$ is H-invariant, $Q = f(z)$ with $f(z) = (\text{unit}) \cdot (z - z(t_2))$. Evaluating at t_1,

$$v_\varpi\big(z(t_1) - z(t_2)\big) = m\, v_\varpi(t_1 - t_2) \in \{4, 3\}.$$

The supplement checks this multiplication of lengths on the involution $t \mapsto -(t+2)/(t+1)$ at $p = 2$. In $K = \mathbf{Q}(\zeta_{12})$ both 2 and 3 have $e_v = f_v = 2$, and the uniformizer of K_v is that of k_p. Summing $\frac{1}{4}$ length $\cdot \log N(v)$ over $v \mid p$ gives the stated bounds.

(c) Each elliptic section meets at most one other at each $v \mid p$, and $a_i a_j \le \frac{4}{9} < 1$. Summing over ordered pairs gives the bound. □

Remark 3.4. It is tempting to identify the deformation coordinate with a coordinate on the coarse space, which would give the length $v(\mathfrak{d})$ on the coarse model. This is incorrect. For the involution above, $t_+ - t_- = 2i$ while $z_+ - z_- = -4i$, which is the factor m in Lemma 3.3(b).

4. The self-intersection of the orbifold canonical bundle

Lemma 4.1 (comparison in codimension one). Let $p \mid M$ and $\pi : \mathcal{X}_M \to \mathcal{X}_{M/p}$. Then, as $\mathbf{Q}$-Weil divisors,

$$L_M = \pi^* L_{M/p} + C_2.$$

Proof. On a normal surface it suffices to compare coefficients at codimension one points.

Horizontal divisors. If $P \mapsto Q$ has ramification index r on the generic fibre, the orbifold orders satisfy $m_Q = r m_P$. The coefficient of P in $L_M - \pi^* L_{M/p}$ is $(r-1) + (1 - 1/m_P) - r(1 - 1/m_Q) = 0$.

Vertical divisors over $q \neq p$. At the generic points of the fibres over q the map π forgets a level structure prime to q, which is étale on fine covers, and coarsening introduces no inertia at these generic points. This is the argument in the proof of [26, Lemma 2.1], which treats both $q \nmid D$ and $q \mid D$ (the non-split case). So the coefficient is 0.

The fibre over p. On C_1 the map π is generically an isomorphism, so the coefficient is 0. At the generic point of C_2, which is ordinary, Serre–Tate coordinates (transported to false elliptic curves by [3, Theorem 4.5, Corollary 4.6]) give $\widehat{\mathcal{O}}_{\mathcal{X}_{M/p}} \to \widehat{\mathcal{O}}_{\mathcal{X}_M}$ in the form $A = \mathbf{Z}_p[[T]]_{(p)} \to B = \mathbf{Z}_p[[t]]_{(p)}$, $1 + T = (1+t)^p$. The residue field extension $\mathbf{F}_p((T)) \subset \mathbf{F}_p((t))$ is generated by t and $e = 1$, so $B = A[t]/(f)$ with $f(t) = (1+t)^p - 1 - T$. The different is $(f'(t)) = (p(1+t)^{p-1})$, whose valuation along C_2 is 1 because C_2 has multiplicity one in the fibre. Hence the coefficient of C_2 is 1. □

Lemma 4.2 (thick nodes). At a point x with local ring $W[[u,v]]/(uv - p^k)$, the two branches $\{u = 0\}$, $\{v = 0\}$ have Mumford intersection number $1/k$.

Proof. The minimal resolution replaces x by a chain $E_1, \dots, E_{k-1}$ of (-2)-curves, with the strict transform $\tilde{C}_1$ meeting E_1 and $\tilde{C}_2$ meeting E_{k-1}. The Mumford pull-back of C_2 is $\tilde{C}_2 + \sum_{i=1}^{k-1} \frac{i}{k} E_i$, which is orthogonal to every E_i. Hence $C_1 \cdot C_2 = \tilde{C}_1 \cdot (\tilde{C}_2 + \sum \frac{i}{k} E_i) = \frac{1}{k}$. □

Lemma 4.3 (mass). Let $p \mid M$ and $d' = d_{M/p}$. Summing over the geometric nodes x of the fibre of $\mathcal{X}_M$ over p, with $k_x = \ell([I_x])$,

$$\sum_x \frac{1}{k_x} = \frac{\varphi(Dp)\psi(M/p)}{12} = \frac{p-1}{2}\, d', \qquad \text{hence} \qquad C_1 \cdot C_2 = \frac{p-1}{2}\, d' \log p.$$

Proof. By Section 2.2 the nodes correspond to the ideal classes of an Eichler order of level M/p in B_{Dp}, and $k_x = \#\mathcal{O}_l(I_x)^\times/\{\pm 1\}$. The Eichler mass formula [22, Theorem 25.3.18] gives $\sum_x 1/k_x = \frac{1}{12}\prod_{q\mid Dp}(q-1)\prod_{\ell\mid M/p}(\ell+1)$. Combine with Lemma 4.2. □

Proposition 4.4 (level-raising recursion). For $p \mid M$,

$$\bar{L}_M^2 = (p+1)\,\bar{L}_{M/p}^2 + d_M\,\frac{3p+1}{2(p+1)}\log p,$$

and consequently

$$\bar{L}_M^2 = \psi(M)\,\bar{L}_1^2 + d_M \sum_{p\mid M}\frac{3p+1}{2(p+1)}\log p. \tag{4.1}$$

Proof. Write $A = \bar{L}_{M/p}$ and F' for the fibre of $\mathcal{X}_{M/p}$ over p; it is reduced and irreducible because $\mathcal{X}_{M/p}$ has good reduction at p. By Lemma 4.1, $\bar{L}_M = \pi^*A + C_2$, with C_2 carrying the trivial metric; at infinity the metric of $\bar{L}_M$ is the pull-back of that of A. Hence

$$\bar{L}_M^2 = (\pi^*A)^2 + 2\,\pi^*A\cdot C_2 + C_2^2.$$

First, $(\pi^*A)^2 = \deg(\pi)A^2 = (p+1)A^2$. Second, $\pi_*C_2 = pF'$ since Frobenius has degree p, so $\pi^*A\cdot C_2 = pA\cdot F' = p\,d'\log p$. Third, the fibre is $C_1 + C_2$ with multiplicities one, so $C_2\cdot(C_1+C_2) = 0$ and $C_2^2 = -C_1\cdot C_2 = -\frac{p-1}{2}d'\log p$ by Lemma 4.3. Adding, the increment is $\big(2p - \frac{p-1}{2}\big)d'\log p = \frac{3p+1}{2}d'\log p$, and $d_M = (p+1)d'$. For the iteration, observe that a later prime $p' \mid M$ is unramified in codimension one at p, so the pull-back of $C_2^{(p)}$ is again reduced; the steps do not interact. □

The supplement checks the identities in Lemma 4.3 and Proposition 4.4 over 1316 triples (D, M, p). It also checks the case $D = 55$, $p = 17$, where four of the 56 superspecial points have $k_x = 3$: $d' = 20/3$, $\sum 1/k_x = 52 + 4/3 = 160/3$, increment $\frac{520}{3}\log 17$. Points with $k_x > 1$ are thus accounted for with weight $1/k_x$ and produce no separate correction.

Corollary 4.5. With $c_* = -\zeta'(-1)/\zeta(-1) - \frac{1}{2}$,

$$\bar{L}_M^2 = d_M\Big[2c_* + \sum_{q\mid D}\frac{3q-1}{2(q-1)}\log q + \sum_{p\mid M}\frac{3p+1}{2(p+1)}\log p\Big].$$

Proof. Yuan's theorem [25, Theorem 1.1] states $\bar{L}_1^2/(2\deg L_1) = c_* + \sum_{q\mid D}\frac{3q-1}{4(q-1)}\log q$ for maximal U, with $\deg L_1 = d_1$. Substitute into (4.1) and use $\psi(M)d_1 = d_M$. □

4.6. Passage to a regular model. Let $\widetilde{\mathcal{X}} \to \mathcal{X}_M \otimes \mathcal{O}_K$ be the minimal resolution. At $p \geq 5$ the extension $K/\mathbf{Q}$ is unramified, and the singular points are the A_{k-1} points of Section 2.2. At $2, 3$ the model is smooth by Lemma 3.3(a), and stays smooth after base change. The resolution of a rational double point is crepant: if $K_{\widetilde{\mathcal{X}}} = \rho^*K + \sum b_iE_i$, adjunction gives $K_{\widetilde{\mathcal{X}}}\cdot E_i = 0 = \rho^*K\cdot E_i$, and negative definiteness forces $b_i = 0$. By Proposition 3.2 the elliptic sections avoid the singular points. Hence the orbifold canonical bundle of $\widetilde{\mathcal{X}}$ is ρ^*L_M, its self-intersection is $\bar{L}_M^2$, and the heights of the σ_i are unchanged. The model $\widetilde{\mathcal{X}}$ is regular and semistable. For $p \mid N$, the number of geometric singular points of a geometric fibre of $\widetilde{\mathcal{X}}$ over $\overline{\mathbf{F}}_p$ is the Artin conductor δ_p; since $p \geq 5$ is unramified in K, dividing $\sum_{v\mid p}\delta_v\log N(v)$ by $[K:\mathbf{Q}] = 4$ gives $\delta_p\log p$ in [7, Theorem 10.1]. This node count is the discriminant term of the arithmetic surface in the Riemann–Roch formula; it is not the conductor exponent of the Jacobian. Each A_{k-1} point contributes $k \geq 1/k$ nodes, so by Lemma 4.3 and the Čerednik–Drinfeld description

$$\delta_p \geq \frac{p-1}{2(p+1)}d_M\ \ (p\mid M), \qquad \delta_q \geq \frac{\varphi(D/q)\psi(Mq)}{12} = \frac{q+1}{2(q-1)}d_M\ \ (q\mid D), \tag{4.2}$$

the second again by the mass formula for $B_{D/q}$ at level Mq.

5. Heights of elliptic sections

Lemma 5.1. For every elliptic section, $h_{\bar{L}_M}(\sigma_i) = \frac{1}{2}\log N + O(1)$. Consequently $|\widehat{\deg}\,\psi_W| = O(e\log N)$, where $\psi_W = \sum_i(1 - m_i^{-2})\psi_{i,W}$ is the metrized psi bundle of [7, Section 10.1.2].

Proof. Iterating Lemma 4.1, $L_M = \pi^* L_1 + \sum_{p\mid M} V_p$, where V_p is a reduced sum of components over p; it is the pull-back of $C_2^{(p)}$. Let k be the CM field of σ_i and $p\mathcal{O}_k = \mathfrak{p}\bar{\mathfrak{p}}$ (Proposition 3.2). The $\Gamma_0(p)$-structure of the CM point is connected at one of the places $\mathfrak{p}, \bar{\mathfrak{p}}$ and étale at the other. Hence σ_i meets V_p transversally at the places of K above exactly one of them, and these carry half of the local degree. So $(\sigma_i \cdot V_p) = \frac{1}{2}\log p$ and

$$h_{\bar{L}_M}(\sigma_i) = h_{\bar{L}_1}(\pi\sigma_i) + \tfrac{1}{2}\log M.$$

The point $\pi\sigma_i$ is the Heegner point of discriminant -4 or -3 on X^D. Pasten's formula [16, Theorem 13.2, (39)] gives $h(P_k) = -\frac{L'}{L}(0,\chi_k) + \frac{1}{2}\log(D/d_k)$, which is $\frac{1}{2}\log D + O(1)$. His height is the Arakelov degree of the Hodge bundle computed on fine levels, which agrees with $h_{\bar{L}_1}$ up to the fixed metric constant [26, Lemma 2.1]. Next, adjunction on the smooth locus gives $\sigma_i^* L_M = \frac{1}{m_i}\sigma_i^*\omega + \sum_{j\neq i} a_j \sigma_i^* \mathcal{O}(\sigma_j)$. The Wolpert metric on $\psi_i = \sigma_i^*\omega$ [7, Definition 4.13] differs from the metric induced by the orbifold hyperbolic metric by a constant depending only on m_i, because the hyperbolic cone of angle $2\pi/m_i$ has a universal model in rotationally symmetric coordinates. Therefore $\widehat{\deg}\,\psi_{i,W} = m_i h_{\bar{L}_M}(\sigma_i) - m_i \sum_{j\neq i} a_j (\sigma_i,\sigma_j)_{\mathrm{fin}} + O(1)$. The cross terms are $O(1)$ by Proposition 3.2 and Lemma 3.3. □

6. Proof of Theorem 1.1

Lemma 6.1 (one-sided spectral bound). Let Γ be a cocompact Fuchsian group with area A and e elliptic points of orders in $\{2,3\}$. Let $Z(s,\Gamma)$ be its Selberg zeta function. Then $\log Z'(1,\Gamma) \le C_0(A+e+1)$ with an absolute constant C_0.

Proof. Let $\Theta(t) = \operatorname{Tr} e^{-t\Delta}$. The Selberg trace formula writes $\Theta = I + E + H$. The identity term is $I(t) = A\,k(t)$ with a universal function $k(t) = \frac{1}{4\pi t} - \frac{1}{12\pi} + O(t)$ as $t \downarrow 0$. The elliptic term $E(t)$ is a sum of at most $2e$ universal functions, one for each elliptic conjugacy class (order $m \in \{2,3\}$, rotation $k \in \{1,\dots,m-1\}$), each smooth on $[0,1]$. The hyperbolic term $H(t) = \sum_\gamma \frac{\ell(\gamma_0)}{2\sinh(\ell(\gamma)/2)} \frac{e^{-t/4-\ell(\gamma)^2/(4t)}}{\sqrt{4\pi t}}$ is non-negative and decays faster than any power as $t \downarrow 0$. Thus $I + E = \frac{a}{t} + b + R(t)$ with $a = A/(4\pi)$, $|b| \le C(A+e)$ and $\int_0^1 |R(t)|\frac{dt}{t} \le C(A+e)$. Splitting the Mellin transform of $\Theta - 1$ at $t = 1$ and differentiating at $s = 0$ gives

$$\log\det{}'\Delta = a - \gamma(b-1) - \int_0^1 R(t)\frac{dt}{t} - \int_0^1 H(t)\frac{dt}{t} - \int_1^\infty (\Theta(t)-1)\frac{dt}{t}.$$

The last two integrands are non-negative, because $H \ge 0$ and $\Theta - 1 = \sum_{\lambda_j>0} e^{-t\lambda_j} \ge 0$. Dropping them gives $\log\det{}'\Delta \le C(A+e+1)$. By [7, Proposition 8.1] with $c = 0$ (there are no model cusps, so the relative zeta function of [7, §8.2] is $\sum_{\lambda_j>0}\lambda_j^{-s}$ and $\det(\Delta_{\mathrm{hyp}},\Delta_a) = \det{}'\Delta$; the setting of [7, §8, §10.1.1] allows $c = 0$), $\log\det{}'\Delta - \log Z'(1,\Gamma)$ is a fixed linear combination of $2g-2+\sum_i(1-m_i^{-1})$ and of the numbers of elliptic points of each type. □

No spectral gap is needed. Small eigenvalues enlarge the discarded integral $\int_1^\infty(\Theta-1)\,dt/t$, which only helps the upper bound.

6.2. Conclusion. The arithmetic Riemann–Roch theorem [7, Theorem 10.1] for $\widetilde{\mathcal{X}} \to \operatorname{Spec}\mathcal{O}_K$ with the sections σ_i reads

$$12\,\widehat{\deg}\,\lambda(\mathcal{O})_Q - \delta + \widehat{\deg}\,\psi_W = \bar{L}_M^2 - \sum_{i\neq j} a_i a_j (\sigma_i,\sigma_j)_{\mathrm{fin}},$$

where $\delta = \sum_p \delta_p \log p$. The Quillen metric is $\|\cdot\|_Q = (C(\Gamma)Z'(1,\Gamma))^{-1/2}\|\cdot\|_{L^2}$, and by Serre duality $\widehat{\deg}\,\lambda(\mathcal{O})_{L^2} = h_F(J) + O(\log A + 1)$. Hence

$$12\,h_F(J) = \bar{L}_M^2 + \delta - \widehat{\deg}\,\psi_W - \sum_{i\neq j} a_i a_j (\sigma_i,\sigma_j)_{\mathrm{fin}} - 6\log Z'(1,\Gamma) + O(g+e). \tag{6.1}$$

We bound each term from below. - By Corollary 4.5 and (4.2), the coefficient of $d_M \log q$ for $q \mid D$ is $\frac{3q-1}{2(q-1)} + \frac{q+1}{2(q-1)} = \frac{2q}{q-1}$; we keep this factor rather than bounding it by 2. The coefficient of $d_M \log p$ for $p \mid M$ is $\frac{3p+1}{2(p+1)} + \frac{p-1}{2(p+1)} = \frac{2p}{p+1}$. So

$$\bar{L}_M^2 + \delta \;\ge\; 2d_M\Big[\sum_{q\mid D}\frac{q}{q-1}\log q + \sum_{p\mid M}\frac{p}{p+1}\log p\Big] - O(d_M).$$

- $-\widehat{\deg}\,\psi_W \geq -O(e\log N)$ by Lemma 5.1. - $-\sum a_i a_j(\sigma_i,\sigma_j)_{\mathrm{fin}} \geq -O(e)$ by Proposition 3.2 and Lemma 3.3(c). - $-6\log Z'(1,\Gamma) \geq -O(g+e)$ by Lemma 6.1, since $A = 2\pi d_M$ in the curvature -1 normalization and $d_M = O(g+e)$.

By (0.1), $d_M \geq 2g-2$, and $e \leq 2^{\omega(N)+1}$. This proves Theorem 1.1. □

Remark 6.3 (what positivity alone gives). Faltings' Noether formula [5, Theorem 6], [14] is $12h_F = \bar{\omega}^2_{\mathrm{Ar}} + \sum_p \delta_p \log p + \delta_F(X(\mathbf{C})) - 4g\log 2\pi$. Combine it with $\bar{\omega}^2_{\mathrm{Ar}} \geq 0$ [5, Theorem 5], with Wilms' bound $\delta_F > -2g\log 2\pi^4$ [24, Corollary 1.2], and with $\delta_p \geq t_p$, the toric rank: $t_q = g$ for $q \mid D$ and $t_p = g\frac{p-1}{p+1} - O(2^{\omega(N)})$ for $p \mid M$. This yields $12h_F \geq g[\log D + \sum_{p\mid M}\frac{p-1}{p+1}\log p] - O(g + 2^{\omega(N)}\log N)$, i.e. the constant 1/12. Theorem 1.1 improves this to 1/3. The improvement comes from evaluating $\bar{L}^2$ instead of using its positivity, together with the mass-weighted node count (4.2).

7. The Petersson-ratio channel for abc

We recall the setting of Pasten [16] (numbering of arXiv:1705.09251v4). Let $a+b=c$ be coprime positive integers, $R = \mathrm{rad}(abc)$, E the Frey curve and N its conductor. For a suitable factorization $N = DM$ with $D > 1$, Pasten bounds $\prod_{p\mid D} v_p(\Delta_E)$ through [16, (54)]:

$$\log \prod_{p\mid D} v_p(\Delta) = 2\log c_f + 2\log\|f\| - 2\log\|f_{D,M}\| + \log D + O_S\Big(\log\log N + \frac{\log D}{\log\log D}\Big),$$

where f is the newform of E, $f_{D,M}$ its integral Jacquet–Langlands transfer to $S_2^D(M)$, and c_f the Manin constant. The denominator is controlled by [16, Theorem 14.1]: $-\log\|h\| \leq (\frac{5}{6}+\varepsilon)\log N + \frac{1}{2}\log M$ for every non-zero integral $h \in S_2^D(M)$. This gives $\prod_{p\mid abc} v_p(abc) \ll_\epsilon R^{8/3+\epsilon}$ [16, Theorem 16.7].

Definition 7.1 (exponent-profile relaxation). For a squarefree R with largest prime factor P, numbers $C \geq C_2 \geq 1$ and a family $\mathcal{V} = \{(D, V_D)\}$ of divisors $D \mid R$ with bounds $V_D \geq 1$, put

$$\mathcal{B}(R; C, C_2, \mathcal{V}) = \max\Big\{\sum_{p\mid R} e_p \log p :\ e_p \in \mathbf{Z}_{\geq 1},\ e_p \leq C,\ e_2 \leq C_2 \text{ if } 2 \mid R,\ \prod_{p\mid D} e_p \leq V_D \text{ for all } (D,V_D)\in\mathcal{V}\Big\}.$$

Since $\log c \leq \sum_{p\mid R} v_p(abc)\log p$, any upper bound for $\log c$ derived only from per-prime bounds $v_p(abc) \leq C$, the bound $v_2(abc) \leq C_2$, and product bounds $\prod_{p\mid D} v_p(abc) \leq V_D$ is at least $\mathcal{B}(R; C, C_2, \mathcal{V})$. We say that such a derivation stays *inside the relaxation model*. The separate bound C_2 is part of the model because Pasten proves $v_2(abc) \ll_\epsilon R^\epsilon$ [16, Lemma 15.2, (56)] by linear forms in logarithms, while the product bounds involve only odd primes. Without it, the profile $e_2 = \lfloor C \rfloor$, $e_p = 1$ ($p \neq 2$) is admissible whenever $2 \mid R$ and all D are odd, and $\mathcal{B}$ is then large for a reason unrelated to the V_D. Apart from this estimate at 2, the model discards the additive relation $a+b=c$ and all further inputs, in particular further linear-forms-in-logarithms estimates. It is an assumption about the method and not a theorem about Pasten's argument.

Lemma 7.2. Assume P is odd, and let $k = \min(\lfloor C\rfloor, \min_{D\ni P}\lfloor V_D\rfloor)$, where the inner minimum over an empty set is $+\infty$. Then $\mathcal{B}(R; C, C_2, \mathcal{V}) \geq \log R + (k-1)\log P$.

Proof. The profile $e_P = k$, $e_p = 1$ for $p \neq P$ is admissible: every product $\prod_{p\mid D} e_p$ is 1 or k, and $e_2 = 1$. □

Proposition 7.3 (lattice ceiling). Let $\mathcal{L} \subset S_2^D(M)$ be Pasten's lattice of integral forms and $\mu(\mathcal{L}) = \min_{0\neq h\in\mathcal{L}}\|h\|^2$. Under the hypotheses of Theorem 1.1, $\mu(\mathcal{L}) \leq N^{1/3+o(1)}$, uniformly in (D, M).

Proof. The Néron cotangent lattice Ω_J of J is contained in $\mathcal{L}$ (via the difference map $X\times X \to J$ on the smooth locus, extended by the Néron property). Its Gram matrix G for the Petersson product satisfies $\det G = \exp(-2h_F(J)+O(g))$; the factor $e^{O(g)}$ accounts for the normalization of the Petersson norm in [16]. Hermite's inequality gives $\mu(\mathcal{L}) \leq \mu(\Omega_J) \leq \gamma_g(\det G)^{1/g}$ with $\gamma_g \leq 1 + g/4$. By Corollary 1.2, $(\det G)^{1/g} \leq N^{-2/3+o(1)}$, and $g \leq N$. □

Corollary 7.4. Let $\mathcal{F}$ be the set of coprime triples $a+b=c$ with $3 \nmid abc$ whose Frey curve has good reduction at 2, i.e. $r = 0$ in [16, Theorem 16.7], and with $\omega(abc)$ large enough that the proof of [16, Theorem 16.7] uses [16, Theorem 16.4]. For such a triple let V_D be any bound obtained by substituting into [16, (54)] an upper bound $U \geq \|f\|^2$ and a lower bound $\beta_{D,M}$ valid for all non-zero integral forms in $S_2^D(M)$, as in [16, Theorem 14.1]. Let $C \geq R^{2/3}$, e.g. $C \asymp R$,

since $\log|\Delta| \ll \max\{1, h(E)\}$ [27, (1.1)] and $h(E) \ll N$ for the semistable Frey curves in $\mathcal{F}$ [27, Corollary 1.3]; any $C \geq R^{2/3}$ suffices, and let C_2 be any bound with $v_2(abc) \leq C_2 \leq C$, e.g. $C_2 = C'_\epsilon R^\epsilon$ from [16, Lemma 15.2]. Then

$$\mathcal{B}(R; C, C_2, \mathcal{V}) \;\geq\; R^{2/3-o(1)}.$$

Consequently no derivation inside the relaxation model with these inputs yields $\log c \ll R^{1/3-\eta}$ for any $\eta > 0$ on $\mathcal{F}$.

Proof. For $r = 0$ and $3 \nmid abc$ we have $\gcd(N, 6) = 1$ and $N = R/2$. The factorizations used in [16, Theorem 16.7] are $(D, M) = (p_1 \cdots p_m, 1)$ for m even, and $(p_1 \cdots p_m/(p_{j_1} p_{j_2} p_{j_3}),\ p_{j_1} p_{j_2} p_{j_3})$ for m odd. All are squarefree with $D > 1$, so Proposition 7.3 applies. Since $\beta_{D,M} \leq \mu(\mathcal{L}) \leq N^{1/3+o(1)}$ and $U \geq \|f\|^2 = N^{1+o(1)}$ (see [16, Section 3]), and since $2\log c_f \geq 0$ and $\log D \geq 0$, every such V_D satisfies $V_D \geq (U/\beta_{D,M})\, N^{-o(1)} \geq N^{2/3-o(1)}$; the error term in [16, (54)] is $N^{o(1)}$. Let P be the largest prime factor of R; since $R = 2N$ with N odd and $\omega(N) \geq 2$, P is odd, and every odd prime of R lies in some D of the list above (for m odd, each p_j is moved into M for only three of the $m \geq 5$ cyclic choices of (j_1, j_2, j_3) in the proof of [16, Theorem 16.7]). Lemma 7.2 applies with $k \geq \min(\lfloor C \rfloor, N^{2/3-o(1)}) = N^{2/3-o(1)}$ and the profile has $e_2 = 1 \leq C_2$, so $\mathcal{B} \geq (k-1)\log P \geq R^{2/3-o(1)}$. □

Remark 7.5 (the obstruction comes from the height). Proposition 7.3 is needed. Suppose instead that every V_D in $\mathcal{V}$ were at most $V = N^{o(1)}$, and that every odd prime of R lies in some D. For an admissible profile, each D contains at most $\log_2 V$ primes with $e_p \geq 2$, and each such prime has $e_p \leq V$. With $|\mathcal{V}| \leq \omega(N)$ divisors D, this gives $\sum_{p|R} e_p \log p \leq \log R + C_2 \log 2 + \omega(N)\, V \log_2 V \log R$. For a fixed $\epsilon > 0$ and the choice $C_2 = C'_\epsilon R^\epsilon$ this is $R^{\epsilon+o(1)}$; since such a choice is available for every $\epsilon > 0$, the model would then yield $\log c \leq R^{\epsilon+o(1)}$ for every $\epsilon > 0$, a subpower bound. Thus in the model with the bound at 2, it is the ceiling $N^{1/3+o(1)}$ of Proposition 7.3 that forces the lower-bound exponent of at least $2/3$ in Corollary 7.4; we do not claim that this exponent is sharp.

7.6. The family is infinite. Take $(a, b, c) = (16t, 1, 16t+1)$ with $t \equiv 1 \pmod 6$. The substitution $x = 4X$, $y = 8Y + 4X$ transforms $y^2 = x(x - 16t)(x + 1)$ into $Y^2 + XY = X^3 - 4tX^2 - tX$. The latter has discriminant $t^2(16t+1)^2$, which is prime to 6 (checked in the supplement). Hence the conductor is $\mathrm{rad}(t(16t+1))$, prime to 6. By the Chinese remainder theorem, t can be chosen divisible by any finite set of primes ≥ 5, so $\omega(abc)$ is unbounded on the family. The smallest member $(16, 1, 17)$ has conductor 17 and does not satisfy the size condition on $\omega(abc)$.

7.7. Scope. Corollary 7.4 says that the Petersson-ratio channel, with the two norms bounded independently and the outputs combined only through Definition 7.1 (including the separate bound at 2), has exponent at least $2/3$ on $\mathcal{F}$. The ceiling in Proposition 7.3 applies to every bound that holds for all integral forms. It says nothing about bounds specific to the eigenform $f_{D,M}$, about correlated estimates for $\|f\|$ and $\|f_{D,M}\|$, or about arguments that use the additive structure. Pasten's subexponential bounds [17, Theorem 1.4] combine the product bound with linear forms in logarithms. They lie outside Definition 7.1 and are not affected. Pasten [27, Theorem 1.1] bounds $h(E) \ll M\varphi(D)\log\ell$, where ℓ is any prime not dividing N, through the congruence modules of f studied in [16], without using [16, (54)] or [16, Theorem 14.1]. This bounds $\log c$ directly rather than through exponent profiles, so it also lies outside Definition 7.1 and is not addressed by Corollary 7.4; for Frey curves it gives $\log c \ll R$, which does not improve the exponent $1/3$ of Stewart–Yu [20].

Verification and supplement

The load-bearing arguments of Sections 3–6 were checked independently by two AI systems; each found local errors, and these are corrected in the present text (Remark 3.4; the global coefficient $\frac{1}{2}\log p$ in Lemma 5.1; the Čerednik–Drinfeld argument in Lemma 4.1). The cited theorem numbers were compared with the sources. The supplement contains a standard-library Python script with exact rational checks of: - the discriminant table of Lemma 3.1; - the identities of Lemma 4.3, Proposition 4.4 and (4.2) over many (D, M, p); - the case $D = 55$, $p = 17$, together with the Eichler class number formula; - the length multiplication of Lemma 3.3 on an explicit involution; - the Frey model of 7.6.

It also contains one finite numerical illustration of the uniformity $\sum_{p|M} \log p/(p+1) = o(\log N)$.

References

- [1] A. Abbes, E. Ullmo, Auto-intersection du dualisant relatif des courbes modulaires $X_0(N)$, J. reine angew. Math. 484 (1997), 1–70.

- [2] D. Banerjee, D. Borah, C. Chaudhuri, Arakelov self-intersection numbers of minimal regular models of modular curves $X_0(p^2)$, Math. Z. 296 (2020), 1287–1329; arXiv:1710.10915.
- [3] K. Buzzard, Integral models of certain Shimura curves, Duke Math. J. 87 (1997), 591–612, doi:10.1215/S0012-7094-97-08719-6.
- [4] B. Conrad, The Keel–Mori theorem via stacks, preprint (2005).
- [5] G. Faltings, Calculus on arithmetic surfaces, Ann. of Math. 119 (1984), 387–424.
- [6] G. Freixas i Montplet, The Jacquet–Langlands correspondence and the arithmetic Riemann–Roch theorem for pointed curves, Int. J. Number Theory 8 (2012), 1–29.
- [7] G. Freixas i Montplet, A.-M. von Pippich, Riemann–Roch isometries in the non-compact orbifold setting, arXiv:1604.00284; J. Eur. Math. Soc. 22 (2020), 3491–3564.
- [8] B. H. Gross, On canonical and quasi-canonical liftings, Invent. Math. 84 (1986), 321–326.
- [9] M. Grados, A.-M. von Pippich, Self-intersection of the relative dualizing sheaf on modular curves $X(N)$, arXiv:2205.11437.
- [10] J. Jorgenson, J. Kramer, Bounds on Faltings's delta function through covers, Ann. of Math. 170 (2009), 1–43.
- [11] U. Kühn, Generalized arithmetic intersection numbers, J. reine angew. Math. 534 (2001), 209–236.
- [12] S. Kudla, M. Rapoport, T. Yang, Modular forms and special cycles on Shimura curves, Ann. of Math. Stud. 161, Princeton (2006).
- [13] H. Mayer, Self-intersection of the relative dualizing sheaf on modular curves $X_1(N)$, J. Théor. Nombres Bordeaux 26 (2014), 111–161.
- [14] L. Moret-Bailly, La formule de Noether pour les surfaces arithmétiques, Invent. Math. 98 (1989), 491–498.
- [15] P. Michel, E. Ullmo, Points de petite hauteur sur les courbes modulaires $X_0(N)$, Invent. Math. 131 (1998), 645–674.
- [16] H. Pasten, Shimura curves and the abc conjecture, J. Number Theory 254 (2024), 214–335; arXiv:1705.09251v4.
- [17] H. Pasten, The largest prime factor of $n^2 + 1$ and improvements on subexponential abc, Invent. Math. 236 (2024), 373–385; arXiv:2312.03566.
- [18] K. Prasanna, Integrality of a ratio of Petersson norms and level-lowering congruences, Ann. of Math. 163 (2006), 901–967.
- [19] J. Stankewicz, Twists of Shimura curves, arXiv:1208.3594.
- [20] C. L. Stewart, K. Yu, On the abc conjecture II, Duke Math. J. 108 (2001), 169–181.
- [21] E. Ullmo, Hauteur de Faltings de quotients de $J_0(N)$, discriminants d'algèbres de Hecke et congruences entre formes modulaires, Amer. J. Math. 122 (2000), 83–115.
- [22] J. Voight, Quaternion Algebras, Grad. Texts in Math. 288, Springer (2021).
- [23] S. Wewers, Canonical and quasi-canonical liftings, in: ARGOS Seminar on Intersections of Modular Correspondences, Astérisque 312 (2007), 67–86.
- [24] R. Wilms, New explicit formulas for Faltings' delta-invariant, Invent. Math. 209 (2017), 481–539.
- [25] X. Yuan, Modular heights of quaternionic Shimura curves, arXiv:2205.13995.
- [26] X. Yuan, Explicit Kodaira–Spencer map over Shimura curves, arXiv:2205.11334.
- [27] H. Pasten, Improved bounds for Szpiro's conjecture, arXiv:2609.17390v1 (2026).